Appendix A

FIELDBUS COMPARISON SPREADSHEET- Note: whilst every effort is made to ensure the accuracy of this spreadsheet, please report any errors or blanks to the author

FIELDBUS	Standard	Continuous Or Digital	Topology	max Segment Length	max Speed	Wire Type	max devices	max data/ PDU	Inter-Operability	Device Plug and Play	Loop Powered	Control in field	On-Line Config	Time Stamp and synch	Obsolescence robustness	Predictive Maint
ASI	EN50295	Continuous and Digital	bus/tree	100m 300m total	167kb/s	2	32	**4 bits**			Optional					
CAN	ISO11898 ISO11519	DIGITAL	bus	500m/125kb/s 40m/1Mb/s	1Mb/s	2	64	8 bytes								
ControlNet	EN50170 IEC61158	DIGITAL	bus/star/ tree	5km 250m/48 nodes	5Mb/s	**coax**	99	510 bytes								
DeviceNet (ODVA)	EN50325	DIGITAL	bus	500m/125kb/s 100m/500kb/s	500kb/s	4	64	8 bytes								
Foundation Fieldbus	IEC61158 EN50170	Continuous and Digital	bus	2000m 9.5km total	31.25kb/s-H1 100Mb/s –HSE	2	240	246 bytes				patented				
FIP	EN50170 EN50254 IEC61158	Continuous and Digital	bus	1000m	2.5Mb/s	2	256	256 bytes			Optional					
INTERBUS-S	EN50254 IEC61158	Continuous and Digital	ring	12.8km	500kb/s	2/8	255	64 bytes								
LON	ANSI 709-1-A-1999	DIGITAL	bus/tree	6.1km/5kb/s	1.2Mb/s	2	2	228 bytes								

Modbus plus	Proprietary	Continuous and Digital	bus	1.8km	1Mb/s	2	32	32 bytes								
P-Net	EN50170 IEC61158	Continuous and Digital	bus/tree	1200m 9.3 km total	76.8kb/s	2	32 masters 125 slav	56 bytes								
PROFIBUS FMS	EN50170	DIGITAL	bus	19.2km/9.6kb/s 200m/500kb/s	500kb/s	2	127	246 bytes								
PROFIBUS DP	IEC61158	DIGITAL	bus	1km/12Mb/s (4 repeater)	12Mb/s	2	127	246 bytes								
PROFIBUS PA	IEC61158		bus	1.9km	93.75kb/s	2	32	246 bytes								
SERCOS	IEC61491	DIGITAL	ring	250m	16Mb/s	2/fiber	245	16 bytes								
Seriplex	Proprietary	DIGITAL	bus	1000 feet	~250kb/s	4	510	32 bytes								
SwiftNet	Proprietary IEC61158	DIGITAL	bus	360m	5Mb/s	2	>1024	896 bytes								

Courtesy of Jim Russell of www.iceweb.com.au